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Why Transformers are better than LSTM and RNN



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Introduction:

In the world of machine learning, transformer models stand out like superheroes among other models. But what makes them so special? Transformers, with their smart architecture, have become the favorites because they're like language wizards — especially in tasks like understanding words, sentences, and even entire stories. They're super at capturing the connections between words, thanks to their attention skills. Now, why are transformers better than their older pals like RNNs and LSTMs? Let's uncover the magic behind why everyone loves these transformer models!

Parallelization:

Transformers allow for parallel processing of input sequences, enabling more efficient training and inference compared to the sequential nature of RNNs and LSTMs. This leads to faster computation and better scalability.

While the RNN and LSTM can't be trained in parallel, To encode the second word in a sentence I need the previously computed hidden states of the first word. So there is a sequence that needs to be maintained.

Long-Range Dependencies:

Transformers use self-attention mechanisms to capture long-range dependencies in the input sequences. This allows them to consider all positions in the input sequence simultaneously. While RNN and LSTM consider the input sequence sequentially.

Contextual Embeddings:

Transformers generate contextual embeddings by considering the entire context of a word in a sequence. This results in more expressive and nuanced representations compared to fixed embeddings used in traditional models.

Bidirectionality:

Transformers process input sequences bidirectionally, considering both left and right contexts. This bidirectional approach helps in capturing richer contextual information, which is especially important for understanding language semantics.

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eg. Consider a sentence "She is reading a fascinating book."

Bidirectional Model:

Forward Process: “She” → “is” → “reading” → “a” → “fascinating” → “book.”

Backward Process: “book” ← “fascinating” ← “a” ← “reading” ← “is” ← “She.”

There are specific LSTMs that offer bidirectionality, but it's not common to use. While RNN are unidirectional.

Self Attention:

The self mechanism in transformers allows the model to focus on different parts of the input sequence when making predictions. This enables capturing dependencies and relationships between words effectively. While In RNNs and LSTMs the information is passed sequentially.

Adaptability to Sequence Length:

Transformers and LSTMs can handle variable-length sequences, making them suitable for tasks where the length of input varies. While in RNNs the input sequence length is fixed.

Conclusion:

Well, imagine you're reading a book: transformers can see the whole story at once, while the older models had to read word by word. It's like transformers have a magic spell that makes them faster and more powerful. Transformer models have achieved state-of-the-art results in various NLP tasks, including machine translation, text summarization, sentiment analysis, and more. Due to their effectiveness and versatility, transformer-based architectures have become the foundation for many subsequent developments in deep learning and NLP.

Transformers

Lstm

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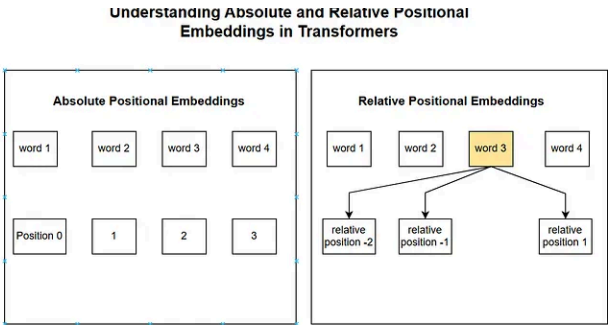
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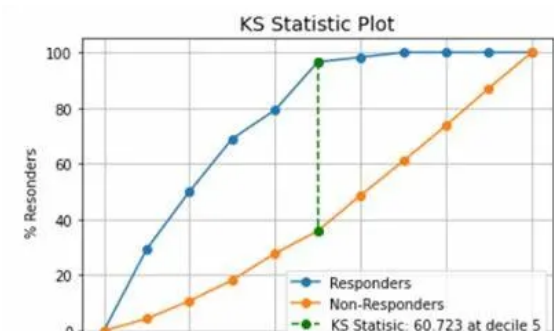
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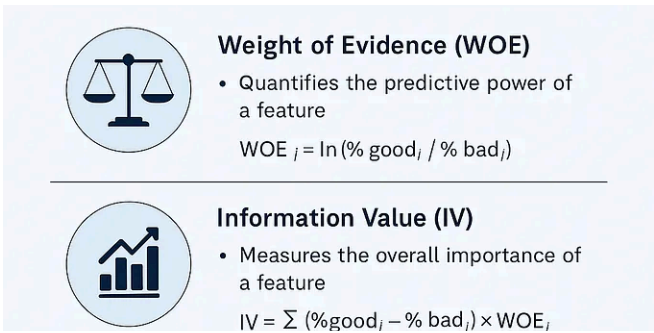
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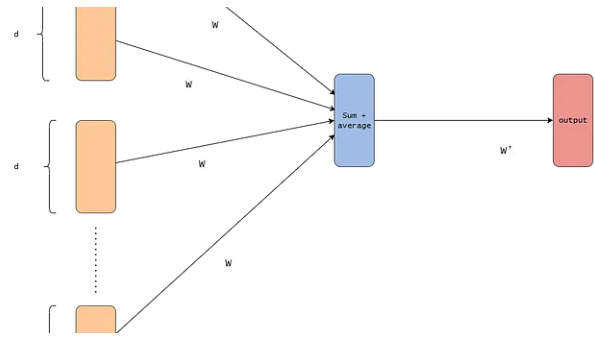
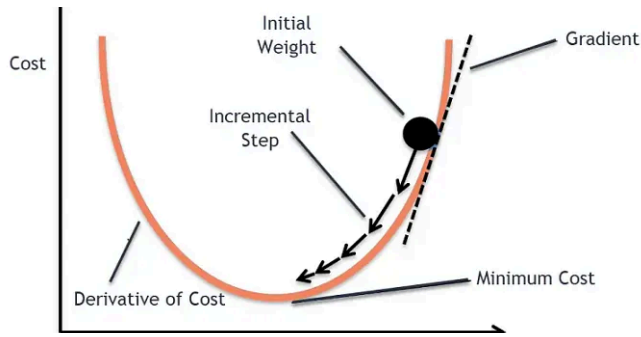
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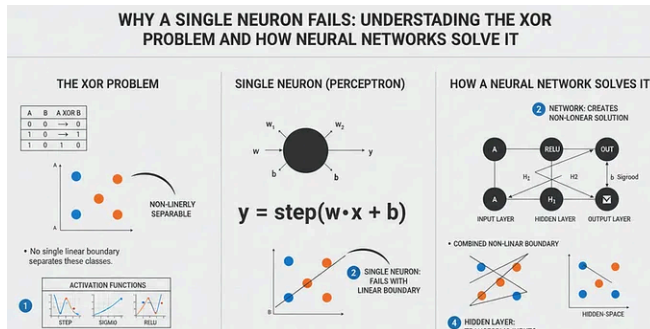


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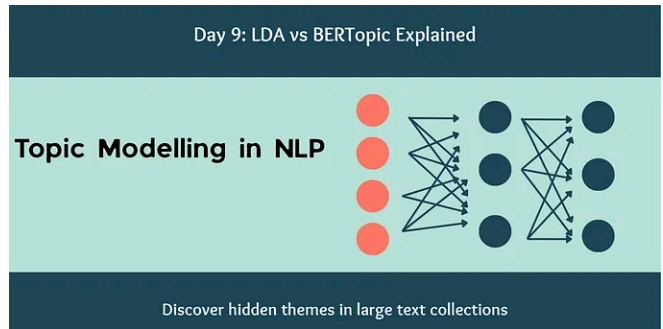


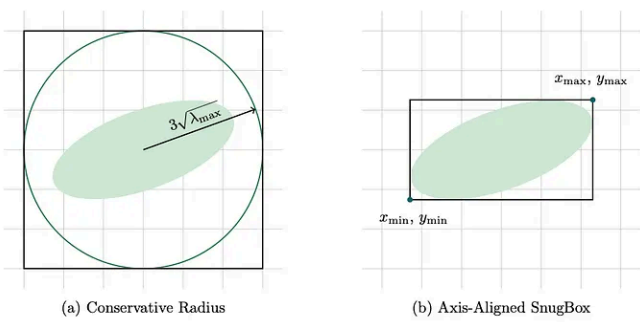
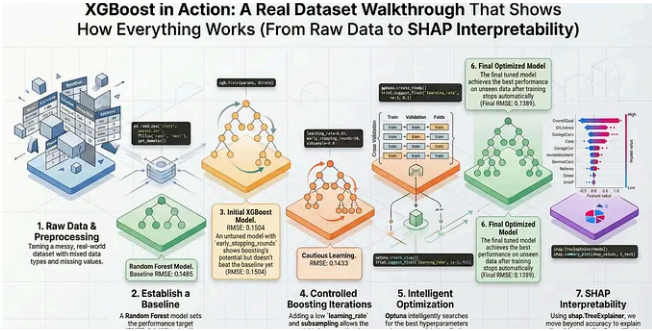
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